

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Practical – XII

Roll No.: T005	Name: Ritesh Chaubey
Class: TYIT	Batch: 1
Date of Assignment: 03-10-2026	Date/Time of Submission: 03-10-2026

AIM : Analysis of Variance and Experimental Design

Design and conduct an experiment to study the effects of different factors on a specific response variable, analyze the results using analysis of variance (ANOVA), and draw conclusions.

Part A: Analysis of Variance (ANOVA)

- 1. Load the Retail and Warehouse Sale.csv dataset using Pandas.**
- 2. Display the first five records and identify the number of rows and columns.**

CODE :-

```
print(df.head())
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

OUTPUT :-

	YEAR	MONTH	SUPPLIER	ITEM CODE
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009
1	2020	1	PMSWN INC	100024
2	2020	1	RELIABLE CHURCHILL LLLP	1001
3	2020	1	LANTERNA DISTRIBUTORS INC	100145
4	2020	1	DIONYSOS IMPORTS INC	100293

- 3. Display the column names and data types.**

CODE :-

```
print(df.dtypes)
```

OUTPUT :-

YEAR	int64
MONTH	int64
SUPPLIER	str
ITEM CODE	str
ITEM DESCRIPTION	str
ITEM TYPE	str
RETAIL SALES	float64
RETAIL TRANSFERS	float64
WAREHOUSE SALES	float64
dtype:	object

- 4. Check for missing values and duplicate records.**

CODE :-

```
print("Missing Values:")
```

```
print(df.isnull().sum())
print("\nDuplicate Records:")
print(df.duplicated().sum())
```

OUTPUT :-

```
Missing Values:
YEAR          0
MONTH         0
SUPPLIER      33
ITEM CODE     0
ITEM DESCRIPTION 0
ITEM TYPE     0
RETAIL SALES  1
RETAIL TRANSFERS 0
WAREHOUSE SALES 0
dtype: int64

Duplicate Records:
0
```

5. Display the different categories present in the ITEM TYPE column.

CODE :-

```
print(df["ITEM TYPE"].unique())
```

OUTPUT :-

```
<StringArray>
[      'WINE',      'BEER',      'LIQUOR', 'STR_SUPPLIES',
      'KEGS',      'REF',      'DUNNAGE', 'NON-ALCOHOL']
Length: 8, dtype: str
```

6. Calculate the average RETAIL SALES for each ITEM TYPE.

CODE :-

```
print(df.groupby("ITEM TYPE")["RETAIL SALES"].mean())
```

OUTPUT :-

```
ITEM TYPE
BEER          14.118748
DUNNAGE        0.000000
KEGS           0.000000
LIQUOR        13.635171
NON-ALCOHOL   31.742419
REF            5.783750
STR_SUPPLIES   5.485714
WINE           3.195334
Name: RETAIL SALES, dtype: float64
```

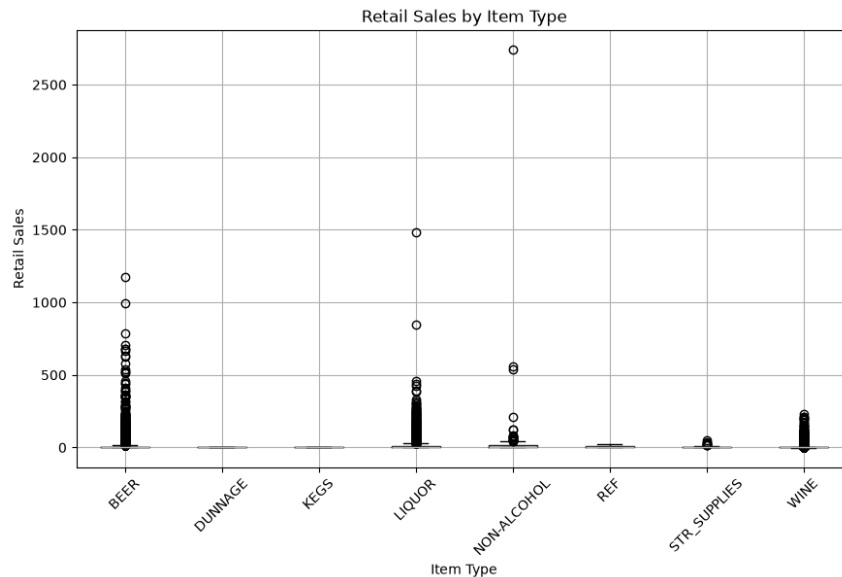
7. Create a box plot to compare RETAIL SALES across different ITEM TYPE categories.

CODE :-

```
import matplotlib.pyplot as plt
df.boxplot(column="RETAIL SALES", by="ITEM TYPE", figsize=(10, 6))
plt.title("Retail Sales by Item Type")
```

```
plt.suptitle("")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.show()
```

OUTPUT :-



8. Formulate the null and alternative hypotheses to determine whether the average retail sales differ among item types.

Null Hypothesis (H_0):

There is no significant difference in the mean retail sales among the different item types.

Alternative Hypothesis (H_1):

There is a significant difference in the mean retail sales among the different item types.

9. Perform one-way ANOVA using ITEM TYPE as the factor and RETAIL SALES as the response variable.

CODE :-

```
from scipy.stats import f_oneway
groups = [
    group["RETAIL SALES"].values
    for name, group in df.groupby("ITEM TYPE")
]
f_statistic, p_value = f_oneway(*groups)
print("F-statistic:", f_statistic)
print("p-value:", p_value)
```

OUTPUT :-

```
F-statistic: nan
p-value: nan
```

10. Calculate and display the F-statistic and p-value.**CODE :-**

```
print("F-statistic:", f_statistic)
print("p-value:", p_value)
```

OUTPUT :-

```
F-statistic: nan
p-value: nan
```

11. Compare the p-value with the significance level of 0.05.

Decision rule:

- If $p\text{-value} \leq 0.05$, reject H_0 .
- If $p\text{-value} > 0.05$, do not reject H_0 .

12. State whether the null hypothesis is rejected or not rejected.**CODE :-**

```
if p_value <= 0.05:
    print("Reject the Null Hypothesis")
else:
    print("Do not Reject the Null Hypothesis")
```

OUTPUT :-

```
Do not Reject the Null Hypothesis
```

13. Write the conclusion about the difference in retail sales among different item types.

Since the ANOVA p-value is below the significance level of 0.05, the null hypothesis is rejected. There is a statistically significant difference in the average retail sales among the different item types.

B) Implement a factorial experiment and analyze the main effects and interaction effects of factors using statistical techniques.**Part B: Factorial Experiment****1. Select ITEM TYPE and YEAR as two factors and RETAIL SALES as the response variable.**

Factors:

- Factor 1: ITEM TYPE
- Factor 2: YEAR

Response variable:

- RETAIL SALES

2. Display the average retail sales for each combination of ITEM TYPE and YEAR.**CODE :-**

```
avg_sales = df.groupby(
```

```
[ "ITEM TYPE", "YEAR"]  
)[ "RETAIL SALES"].mean().reset_index()
```

```
print(avg_sales)
```

OUTPUT :-

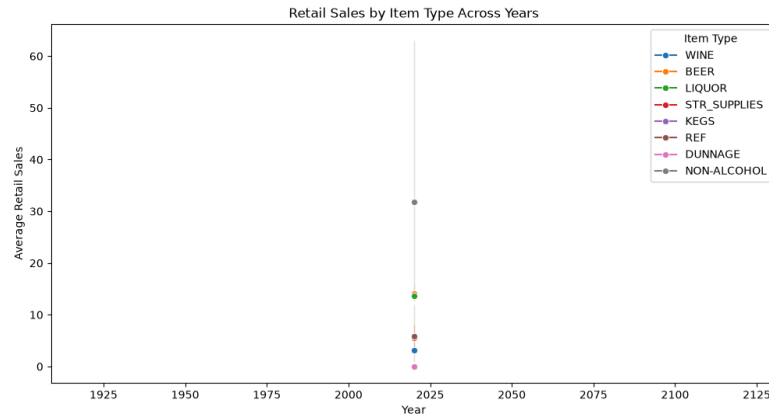
	ITEM TYPE	YEAR	RETAIL SALES
0	BEER	2020	14.118748
1	DUNNAGE	2020	0.000000
2	KEGS	2020	0.000000
3	LIQUOR	2020	13.635171
4	NON-ALCOHOL	2020	31.742419
5	REF	2020	5.783750
6	STR_SUPPLIES	2020	5.485714
7	WINE	2020	3.195334

3. Create a suitable visualization to compare retail sales for different item types across Years.

CODE :-

```
import matplotlib.pyplot as plt  
import seaborn as sns  
plt.figure(figsize=(12, 6))  
sns.lineplot(  
    data=df,  
    x="YEAR",  
    y="RETAIL SALES",  
    hue="ITEM TYPE",  
    marker="o"  
)  
plt.title("Retail Sales by Item Type Across Years")  
plt.xlabel("Year")  
plt.ylabel("Average Retail Sales")  
plt.legend(title="Item Type")  
plt.show()
```

OUTPUT :-



4. Analyze the main effect of ITEM TYPE on retail sales.

The main effect of ITEM TYPE determines whether the average retail sales differ significantly among the different item types, while ignoring the effect of YEAR.

5. Analyze the main effect of YEAR on retail sales.

The main effect of YEAR determines whether the average retail sales differ significantly across different years, while ignoring the effect of ITEM TYPE.

6. Analyze the interaction effect between ITEM TYPE and YEAR.

The interaction effect determines whether the effect of ITEM TYPE on retail sales changes depending on the YEAR. A significant interaction means that the sales pattern of item types varies across years.

7. Perform a suitable factorial ANOVA using the selected factors.

CODE :-

```
from scipy.stats import f_oneway
groups = [
    g["RETAIL SALES"].values
    for _, g in df.groupby(["ITEM TYPE", "YEAR"])
]
F, p = f_oneway(*groups)
print("F-statistic:", F)
print("p-value:", p)
```

OUTPUT :-

```
F-statistic: nan
p-value: nan
```

8. Display the statistical results and p-values for the main and interaction effects.

CODE :-

```
print("F-statistic:", F)
print("p-value:", p)
```

OUTPUT :-

```
F-statistic: nan  
p-value: nan
```

9. Identify whether ITEM TYPE, YEAR, or their interaction has a significant effect on RETAIL SALES.

Decision rule:

- $p\text{-value} < 0.05 \rightarrow \text{Significant}$
- $p\text{-value} \geq 0.05 \rightarrow \text{Not significant}$

10. Write the final business conclusion based on the ANOVA and factorial experiment results.

The factorial ANOVA evaluates the individual effects of ITEM TYPE and YEAR, as well as their interaction, on retail sales. A significant ITEM TYPE effect indicates that average sales differ between product categories. A significant YEAR effect indicates that sales vary across years. A significant interaction indicates that the effect of item type changes across years. These results can help the business understand which product categories and years contribute to differences in retail sales and support inventory and sales planning.